Single Cell Analysis

Cell Ranger Alignment

A. CellBender
- 00_cellBender_adjustMatricies.ipynb

B. Scanpy Pre-processing
- 01_PMACSscanpyPipeline_cellBender.ipynb (SFig. 1a)
- 02a_PMACS_annotation_FetalOnly_final.ipynb (Fig. 1a-c, SFig. 1b,c, SFig. 2a)

C. Partitioning UB/NPC linages with CellRank

-v03a2_scVelo_fetal_nephronAndUB_final.ipynb (SFig. 1d)
-v05a2_cellRank_absorbtionProbability_NephronWithUB.ipynb (SFig. 1f)

D. Nephron and UB subclustering

-02b_PMACS_annotation_FetalOnly_final_Nephron_UB2.ipynb
-02b_PMACS_annotation_FetalOnly_final_Nephron_UB.ipynb

E. NPC lineage probability estimation

-v03b_scVelo_fetal_nephron_final.ipynb (SFig. 3a)
-v04b_fetal_nephrons_randomWalk.ipynb (SFig. 4b,d)
-v05b_cellRank_absorbtionProbability_nephron.ipynb (SFig4, e-h, Fig. 2d)
-v06_selection.ipynb (Fig. 2d)
-v06_self_renew.ipynb (Fig. 2f)
-v06_specification.ipynb (Fig. 2b, SFig. 2e,f).

F. Psudobulk DEG calculation and pathway analysis

-gAgeDEseq.ipynb (SFig. 2c)
-UpsetPlot_fetalTime.ipynb (SFig. 2d)
-GSEApy_fetalTime.ipynb (Fig. 2d, SFig. 2e, SFig. 5)
-CompareDEG_renew.ipynb (Fig. 2g,h)

G. STAR SOLO matrix generation
- v00_StarSolo_final.ipynb

H. Velocity Estimation with scVelo
- v01_import_scVelo_fetal_final.ipynb
- v02_scVelo_fetal_final.ipynb (SFig 1d)

I. UB linage probability estimation

-v03b_scVelo_fetal_UB_final.ipynb
-v05b_cellRank_absorbtionProbability_UB.ipynb (SFig. 4)

Spatial Analysis and integration with Single Cell Data

J. Import CosMx data into SquidPy

-Example_prepareData_flatFiles_images.ipynb
-Example_PartitionSamples.ipynb (SFig. 6a)
-FetalProject_s03_MergeBasicPreprocessing.ipynb (SFig. 6b)
-FetalProject_s03b_PrepareSCVI.ipynb

K. Integration with scRNAseq and imputation

-SCVI_integrate_fetal_2.ipynb
-FetalProject_s04a_postSCVI_preSCANVI_model2.ipynb
-FetalProject_s04b_postSCANVI_post_process_model2.ipynb (SFig. 8a-e, SFig. 12, Fig. 4g)
-SCANVI_integrate_fetal_2_final.ipynb (Fig. 3a, SFig. 7b)
-FetalProject_s04c_postSCANVI_call_spatial_Neighbors_model2.ipynb
-FetalProject_s05_postSCANVI_nephronTranj_model2.ipynb
-FetalProject_s06_post2SCANVI_impute_allGenes_model2.ipynb (SFig. 7d)

L. Examination of Histologic Neighborhoods

Overlay_fetal_FK1.ipynb
FetalProject_s07_analyze_manual_neighborhoods.ipynb (SFig. 11c)
FetalProject_s08_neighborAnalysis_model2.ipynb (SFig. 11b)

M. CytoSignal interaction estimation

-FetalProject_s09_cytoSingal_Export.ipynb
-prepare_FetalCosmx_example_raw.R
-runCytoSignal_sample_imputed.R
-prepareForPythonImport_all_interactions_all_samples_new.R
-FindCommonInts.R

N.Downstream ligand/cell-cell interaction analysis

-FetalProject_s10_analyze_ligand_score_all.ipynb
-FetalProject_s11_analyze_ligand_score_conserved_plots2_new.ipynb (Fig. 5a)
-FetalProject_s11_analyze_ligand_score_corr.ipynb
-FetalProject_s10_analyze_ligand_score_conserved.ipynb
-FetalProject_s10_analyze_ligand_score_conserved_plots.ipynb (Fig 6a-c)
-FetalProject_s11_analyze_ligand_score_corr_PT.ipynb (Fig. 5d)
-FetalProject_s11_analyze_ligand_score_pyGAM.ipynb (Fig. 5c)
-Correlation_Histograms.ipynb (Fig. 5b)
-FetalProject_s12_neighborAnalysis_ligand.ipynb (Fig. 6d)
-FetalProject_s13_analyze_environment_neighborhoods (Fig. 6c, SFig. 15d-f)

O. Other

-example_FOV_plotting.ipynb (Fig. 3b-e, Sfig7g, SFig. 9b, SFig. 10)
-Sankey_cosMxPops.ipynb (SFig. 8b)